Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Cross-Site Request Forgery (CSRF)

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

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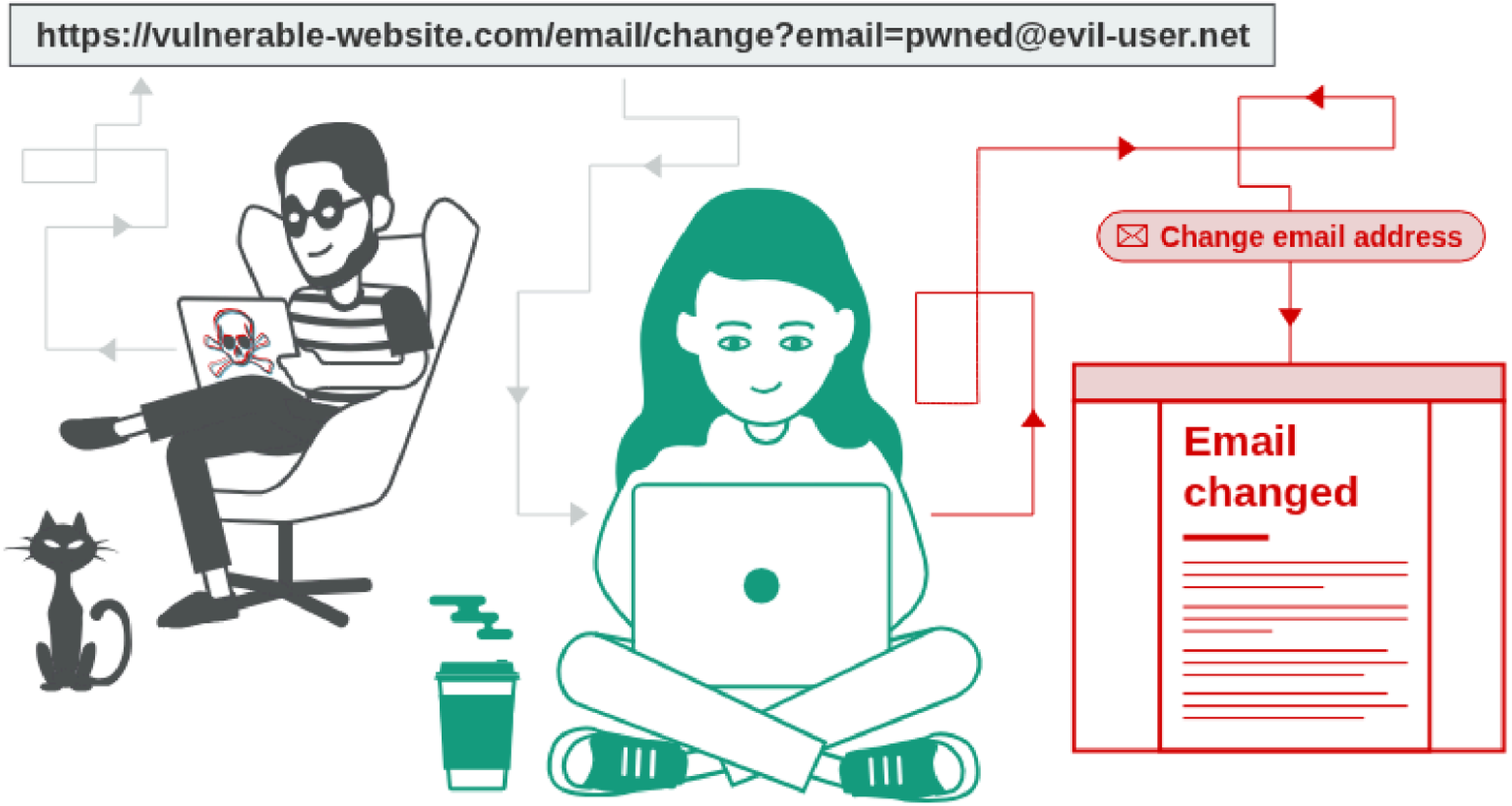
CSRF attacks abuse the **automatic attachment of cookies** to requests done by browsers in order to perform arbitrary actions within the session established by the victim with the target website.

Depending on the nature of the action, the attacker might be able to gain **full control** over the user's account. If the compromised user has a privileged role, the attacker might take full control of the application.



OWASP > A01:2021 – Broken Access Control > CSRF

* Image source: https://portswigger.net/web-security/csrf

Three Conditions for CSRF

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3. **No unpredictable request parameters**: the requests that perform the action do not contain any parameters whose values the attacker cannot determine or guess.

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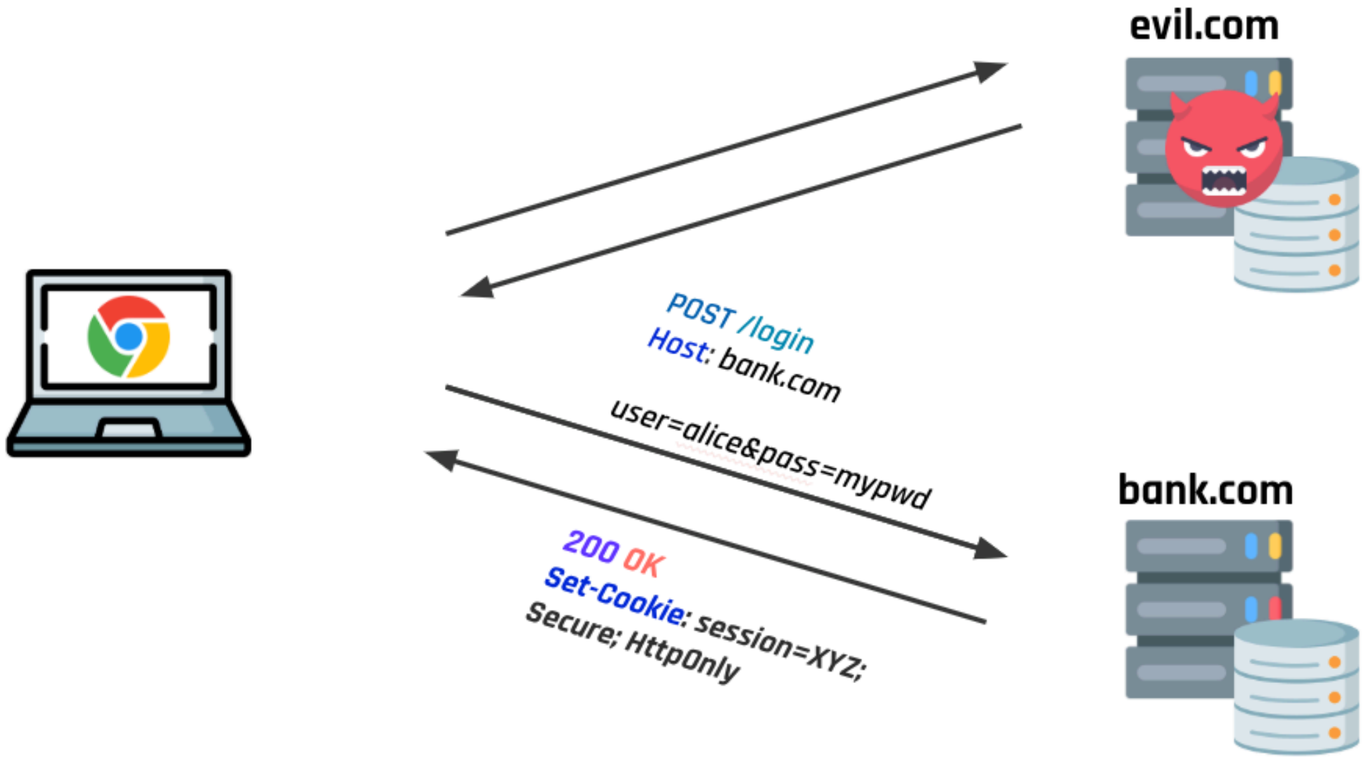
3. The **session cookie is automatically attached** by the browser!

> The target website **cannot distinguish** legitimate requests from those triggered by a third-party website.

CSRF Example (1): Victim Logs In

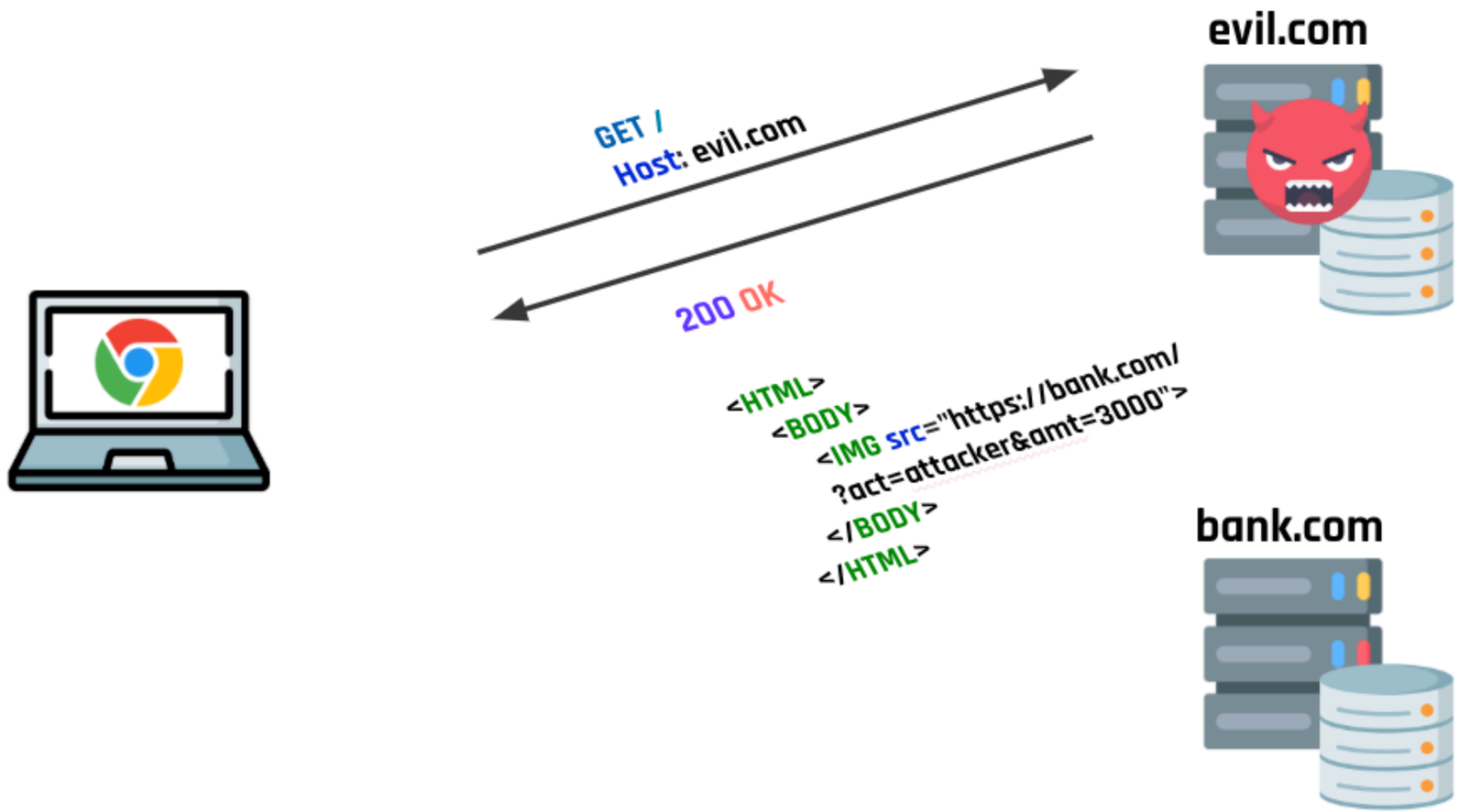
Alice authenticates with `bank.com`:



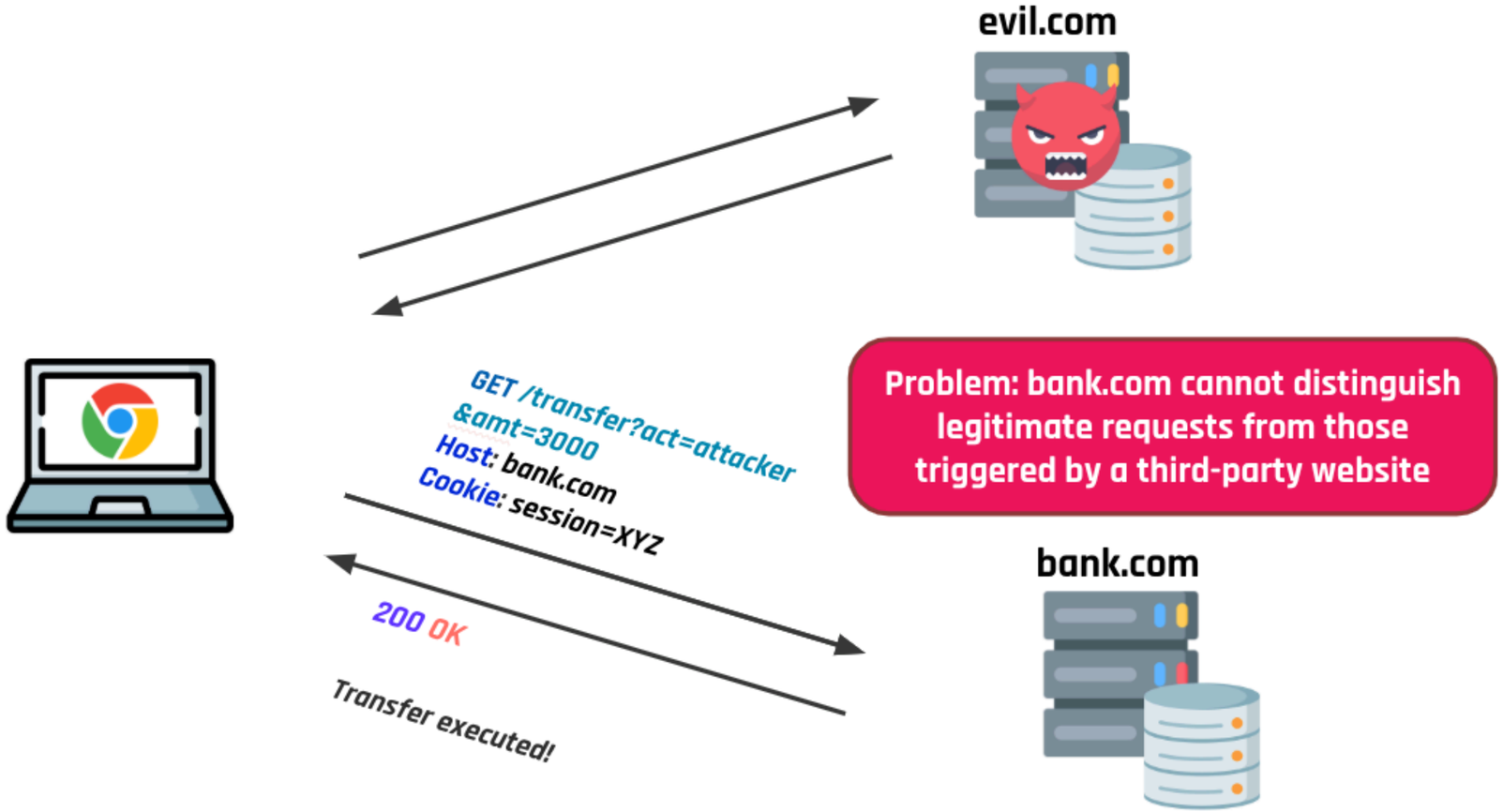
Alice is now **authenticated** – her browser will attach `session=XYZ` to all subsequent requests to `bank.com`.

CSRF Example (2): The Malicious Page

Alice visits `evil.com`, which serves:



CSRF Example (3): The Malicious Request



CSRF on GET Requests

If the sensitive action uses **GET**, the attack is trivial – just embed a URL:

As a hidden image (zero-size):

```
<img src="http://bank.com/transfer?acct=MARIA&amount=100000"
     width="0" height="0" border="0">
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> **NOTE**: The victim does not see anything suspicious – the browser silently makes the request.

CSRF on POST Requests

For **POST** requests, the attacker uses a hidden form with auto-submit:

```html
<html>
  <body onload="document.forms[0].submit()">
    <form action="https://bank.com/transfer" method="POST">
      <input type="hidden" name="acct" value="MARIA" />
      <input type="hidden" name="amount" value="100000" />
    </form>
  </body>
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</html>
```

When the victim loads this page, the form is **automatically submitted** and the browser attaches the session cookie.

CSRF on Other HTTP Methods (PUT, DELETE)

Modern APIs often use PUT or DELETE. An attacker can use JavaScript:

```
<script>
function put() {
    var x = new XMLHttpRequest();
    x.open("PUT", "http://bank.com/transfer", true);
    x.setRequestHeader("Content-Type", "application/json");
    x.send(JSON.stringify({"acct": "BOB", "amount": 100}));
}
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> Unless the target explicitly allows cross-origin requests via CORS: `http Access-Control-Allow-Origin: *`

How to Deliver a CSRF Exploit

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- **Planting on popular websites**: e.g., embedding the exploit in a user comment on a popular site and waiting for victims to visit
- **Self-contained URL**: if the action uses GET, the attacker can directly feed a malicious URL on the vulnerable domain (no external site needed)

CSRF Defenses

Common Defenses Against CSRF

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2. **SameSite Cookies** – a browser mechanism that restricts when cookies are sent in cross-site requests
3. **Referer Validation** – checking the HTTP Referer header to verify the request origin
4. **Fetch Metadata** – using browser-provided headers to detect suspicious request contexts

Anti-CSRF Tokens: Synchronizer Token Pattern

A secret, randomly generated string is embedded by the web application in all HTML forms as a **hidden input field**:

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<form action="/transfer" method="POST">
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Upon form submission, the application checks whether the request contains the expected token. If not, the sensitive operation is **not performed**.

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> **NOTE**: The attacker cannot read the token from the page (blocked by the Same-Origin Policy) and therefore cannot forge a valid request.

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> **NOTE**: This works because a cross-origin attacker cannot read cookies from the target domain (Same-Origin Policy) and therefore cannot set the custom header.

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Most modern web frameworks use anti-CSRF tokens **by default**!

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> Since `Lax` is now the **default**, sites are protected against classic CSRF attacks out of the box – but **no protection** is given against related-domain attackers (requests from a sibling subdomain are same-site).

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The web application can inspect the `Referer` header to verify that the request comes from an allowed page/domain.

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> Referer-based validation is generally **less effective** than CSRF token validation.

Fetch Metadata

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For privacy reasons, these headers are sent only over **HTTPS**.

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- **`Sec-Fetch-User`**: indicates whether the request was triggered by a user action

Fetch Metadata: Sample Policies

Resource isolation policy (mitigates CSRF, XSSI, timing side-channels):

Reject if:

```
Sec-Fetch-Site == 'cross-site'
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Navigation isolation policy (mitigates clickjacking and reflected XSS):

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- **HTTPS alone**: does nothing to defend against CSRF (but should be a prerequisite for any defense)
- **Only validating the Referer header**: can be spoofed or suppressed, leading to false positives or bypasses

Challenge ItIsASecret

ItIsASecret

Description: You are not authorized... Are you?

URL: https://web11.webhack.it

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- **`/admin`**: There is an admin panel with a grant functionality
- **`/admin/grant`**: Only admin can use this feature – it grants access to a user given their UID
- **`/report`**: A page where we can report "broken pages" – the admin will visit the reported URL

Analysis

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Analysis

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- If we submit a URL on a server that **we control**, we can observe that the admin is visiting it – even if our URL is **not** from the subdomain of the challenge!

What if we can make the admin grant access to us?